- Use **test_data** to generate predictions on rows, for outputs to be scored to see how well the model generalises unseen encounters

Challenge #2

" How might AI help air traffic controllers, pilots and ground operations staff make faster and better decisions? "

Description

Today, when a runway closes, weather delays occur or ground equipment fail unexpectedly, communications are reliant on human phone calls, manual software updates etc., leading to cascading delays, excessive holding patterns, and excessive fuel consumption. The future of Air Traffic Management relies on AI assistants that can autonomously communicate and negotiate trade-offs across organisational boundaries.

Want to see

A **Multi-Agent Communication Framework** where ATC AI agents, Pilot AI agents, and ground ops agents are able to negotiate resolutions to real-time disruptions without human intervention until a consensus is reached.

ATC agent → must take care of runway wake separation, taxiway blockage, maximise sector throughput

Pilot Agent → Preserve fuel reserves, ensure flight safety, minimise turbulence and protect passengers' connections

Ground Agent → Optimise gate utilisation, avoid ground-crew overtime, minimise turnaround delays

These 3 agents must exchange structured data, evaluate cross-domain constraints, and converge on an agreed solution. This includes sending **structured messages**, **explicit constraint checking, traceable consensus log**, **deterministic non-AI fallback**

What to avoid

- Combining 3 roles into a single prompt call; The point is to test multi-agent coordination across independent domains
- LLM agents talking to each other in endless, unconstrained loops with no convergence logic / turn limits
- Disregard of safety margins; AI agents should not be allowed to accept solutions that ignore operational constraints

Provided data

US Department of Transportation data for 2015 flight delays and cancellations

- File in *Challenge #2\DATA*
- Download this, pick any row with WEATHER_DELAY value of > 15, or LATE_AIRCRAFT_DELAY > 20
- Pass the parameters in that row into the 3 AI Agents

- Demonstrate how agents negotiate **revised arrival time, gate assignment, or flight plan adjustment** in under 3 turns

Challenge #3

" How might we optimise the SAF supply chain in Kalimantan? "

Description

There is a need to aggressively scale Sustainable Aviation Fuel (SAF) adoption to meet decarbonisation goals. Kalimantan (Indonesian Borneo) possesses abundant agricultural and forestry residues, making it a potential hub for regional SAF production. However, transporting raw biomass, establishing infrastructure, and delivering finished SAF to international airports requires strategic decision-making, and often comes with complex logistical and capital trade-offs.

Want to see

- A **SAF supply chain optimisation model** across Kalimantan that minimises the total cost (OpEx + CapEx + Logistics), while meeting SAF demand at Kalimantan's 4 international airports (Supadio Intl, Iskandar Intl, Sultan Aji Muhammad Sulaiman Sepinggan Intl, Juwata Intl)
- Should include optimal coordinates of depots & refineries
- Facility sizing decisions
- Material flow network
- Infrastructure placement: Deciding the spatial locations and sizes (Small vs Large) for pre-processing depots and SAF refineries
- Optimal routing of raw feedstock to depots, depots to refineries, finally refineries to airports

Feedstock quantity - **6800kt**

Facility Type	Size	Cost (USD)	Production Capacitykilotonne
Depot	Small	2.25M	150 kt
Depot	Large	4.5M	300 kt
Refinery	Small	480M	500 kt
Refinery	Large	765M	1087 kt

Truck Size	Farm to Depot	Cost (USD)tonne/km	Depot to Refinery	Cost (USD)tonne/km	Refinery to Airport	Cost(USD)tonne/km
Small	3 t	0.10/t/km	18 t	0.12/t/km	16 t	0.08/t/km
Large	9 t	0.10/t/km	30 t	0.12/t/km	32 t	0.08/t/km

Data provided

Judging Rubric

AI & Aerospace Technology Track

Six criteria. Score each from 1 to 5; apply the weights below. Total weight: 100%.

Problem realism 20%	Technical soundness 25%	Honest evaluation 20%
Trust and traceability 20%	Impact 10%	Time discipline 5%

Problem realism

20% | 1-5

Does this solve a specific aerospace workflow problem?

Name the user, current process and cost in time, money or risk. Explain the constraints of the chosen challenge, rather than describing aviation in general.

STRONG
A credible user, narrow workflow and clear status quo.

WEAK
A generic aviation chatbot or a solution looking for a problem.

Technical soundness

25% | 1-5

Does the prototype work, with the right method for each task?

Show an end-to-end build. Use LLMs for reading and explanation; use solvers, rules or statistics for scheduling, constraints and detection. Test beyond a scripted demo.

STRONG
Working inputs and outputs; explicit constraints and a sound LLM / non-LLM split.

WEAK
LLM-only legality checks, opaque risk scores or a hard-coded demo.

Honest evaluation

20% | 1-5

What was tested, against what baseline, and where does it fail?

State metrics, test data and results. Label synthetic data and its limits. Show one case where the system is wrong, uncertain or declines, and explain how the user knows.

STRONG
A baseline, measured results and a visible failure with a reason.

WEAK
Only success cases, unsupported accuracy claims or hidden data limits.

Judging Rubric

AI & Aerospace Technology Track | Criteria and scoring guidance

Trust and traceability

20% | 1-5

Can a qualified person verify the output before acting?

Trace claims to a document, revision and section, a timestamped signal or an applied rule. Handle applicability and missing evidence. Keep the qualified person in control.

STRONG
Checkable sources, visible uncertainty and a clear refusal when needed.

WEAK
Confident unsourced answers or a claim to replace expert judgement.

Impact

10% | 1-5

What improves over the current process, and by how much?

Compare against the stated baseline: delay minutes, supplier impact, retrieval quality or false alarms, as relevant. Distinguish measured improvement from estimates.

STRONG
A credible before/after measure and realistic adoption constraints.

WEAK
Large benefit claims with no baseline or unstated assumptions.

Time discipline

5% | 1-5

Was the project submitted early and presented efficiently?

Earlier valid submission is better, using the recorded submission time. Also assess a clear, focused presentation completed within 3 minutes. Give one combined 1-5 score.

STRONG
Earlier submission and a focused, complete presentation within 3 minutes.

WEAK
Later submission and a rushed, unfocused or over-time presentation.

SCORING AND JUDGES' NOTES

Scale: 1 = weak; 2 = limited; 3 = credible; 4 = strong; 5 = exceptional evidence.

Weighted total / 100 = sum of (criterion score / 5 x criterion weight).
Example: technical soundness scored 4/5 earns 20 of its 25 points.

Score criteria separately. Ask: "Where does this fail, and how would your user know?"
Reward sound restraint, not presentation polish. Each project is evaluated by 3 judges.
Time is only 5%: earlier submission does not substitute for quality or evidence.